

The background of the slide is a dark, semi-transparent image of a laptop screen. On the screen, there is a line graph with a blue line and a pie chart with green and blue segments. The text is overlaid on this image.

Case Study #1:

How Can a Wellness Technology Company Play It Smart?

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Executive Summary

Goal

Use Fitbit data to uncover how women engage with smart wellness devices

Outcome

Actionable insights for product and marketing strategy

Top Insight

Usage is high, but drop-off and inconsistent sleep tracking present key re-engagement opportunities



Business Context

- Bellabeat creates health-focused smart products for women
- Chief Creative Officer requested analysis of Fitbit data
- Focus: Sleep, movement, device usage patterns

Business Task

- **Key Question:**

How do women use smart wellness devices, and how can this data drive marketing?

- **Objectives:**

- Identify patterns in device use
- Segment user behaviors
- Recommend creative, targeted strategies

Key Stakeholders



Co-founder and Chief
Creative Officer

Will use the findings to refine marketing strategies.



Co-founder

Interested in how data supports the company's business goals.



Marketing analytics
team

Will interpret and present findings to other departments. Supports the marketing strategy with data insights.

Business Task

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Data Source

- **Source:** Public Fitbit dataset (Kaggle, 2016)
- **Period:** March 12 – May 12, 2016
- **Files:** 29 Excel files from Fitabase folders
- **Data Segments:** Daily, hourly, minute-level data
- **Key Variable:** Participant ID (relational)

Data Limitations

- Old dataset (2016)
- Small sample size (~30 users)
- Self-selected MTurk participants (not fully representative)
- Missing/incomplete variables (e.g., sleep tracking)



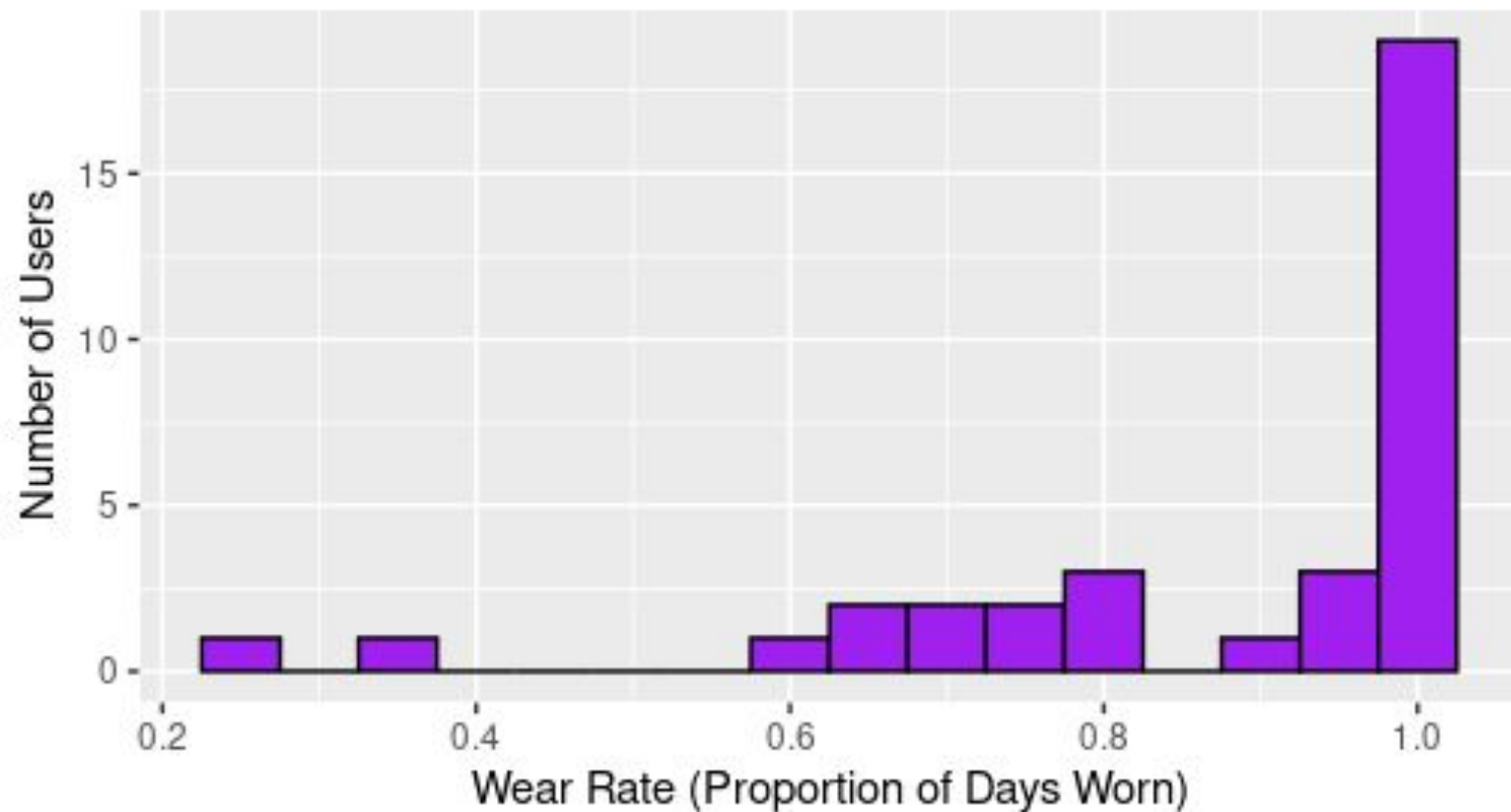
How do women use smart wellness devices?

- Do women wear the device on a daily basis? And consistently over time?
- Is there a drop-off pattern in usage?
- Do women track movement throughout the day?
- Do women wear the device during sleep? How often?
- Do women monitor sleep routines?
- Do usage patterns change on weekends?
- Do women work out using the device? Consistently over time?
- Do women wear the device only or mostly during workouts?

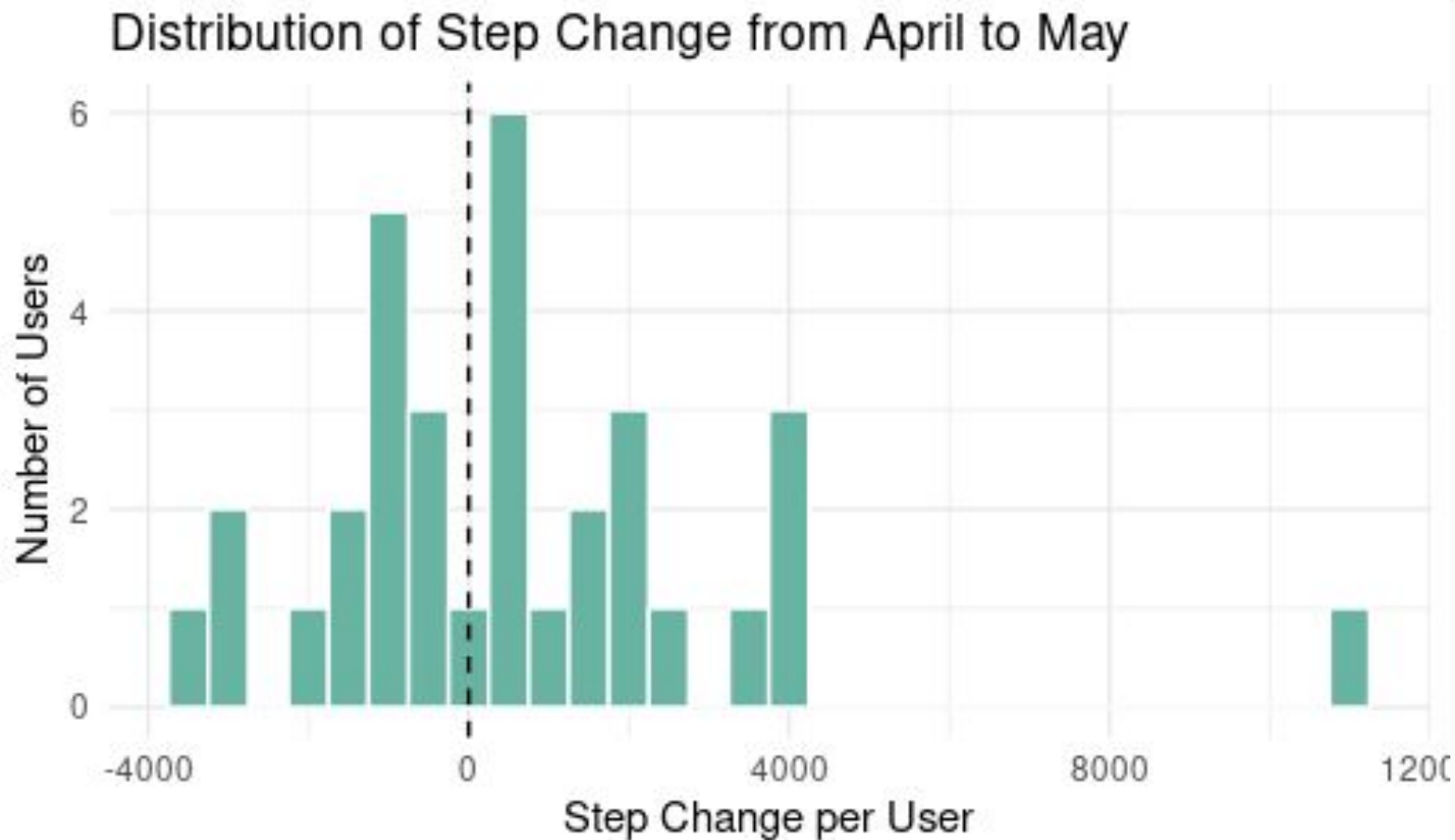
Behavioral Insights

Do women wear the device on a daily basis?

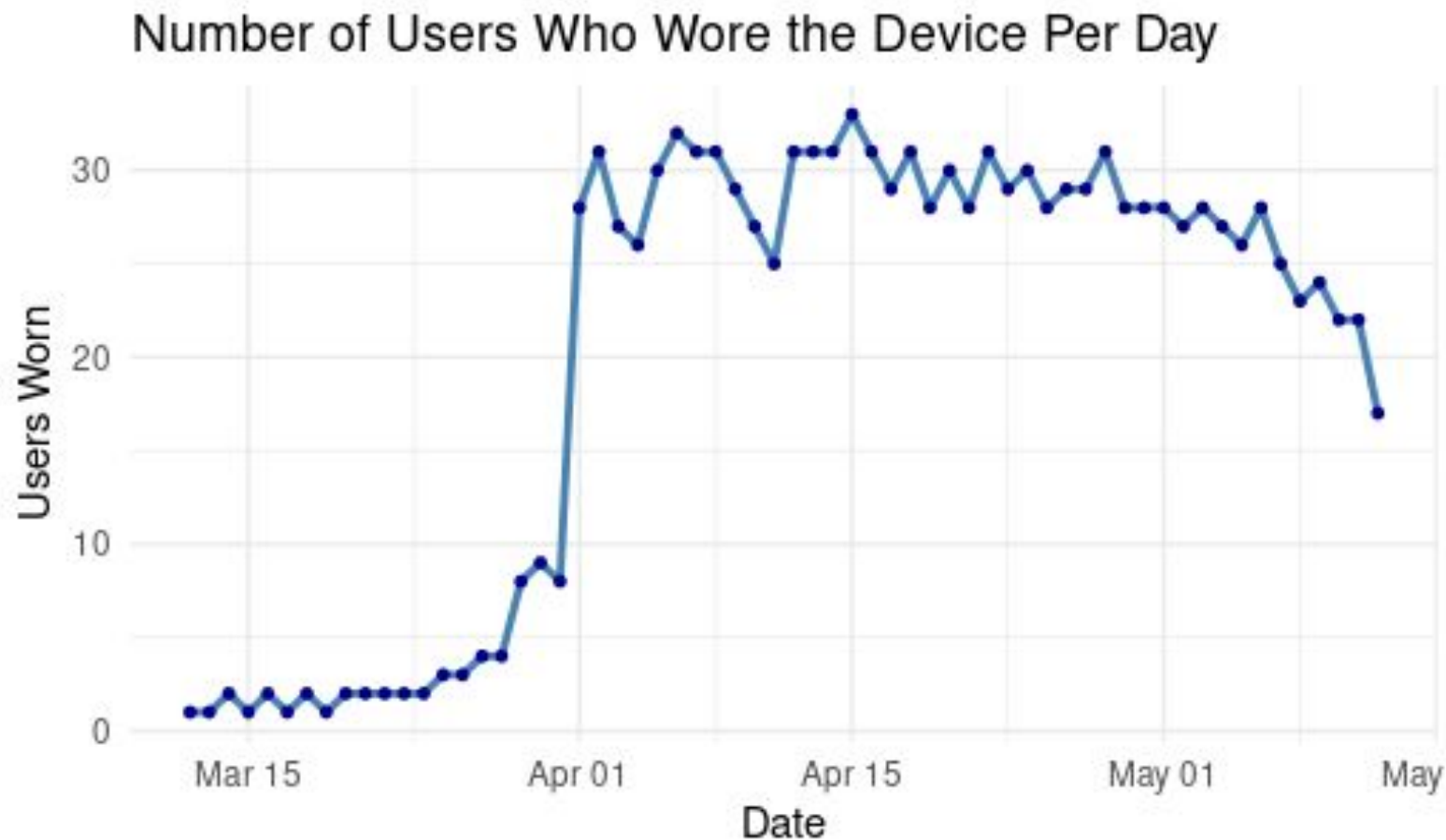
Distribution of Device Wear Rate Across Users



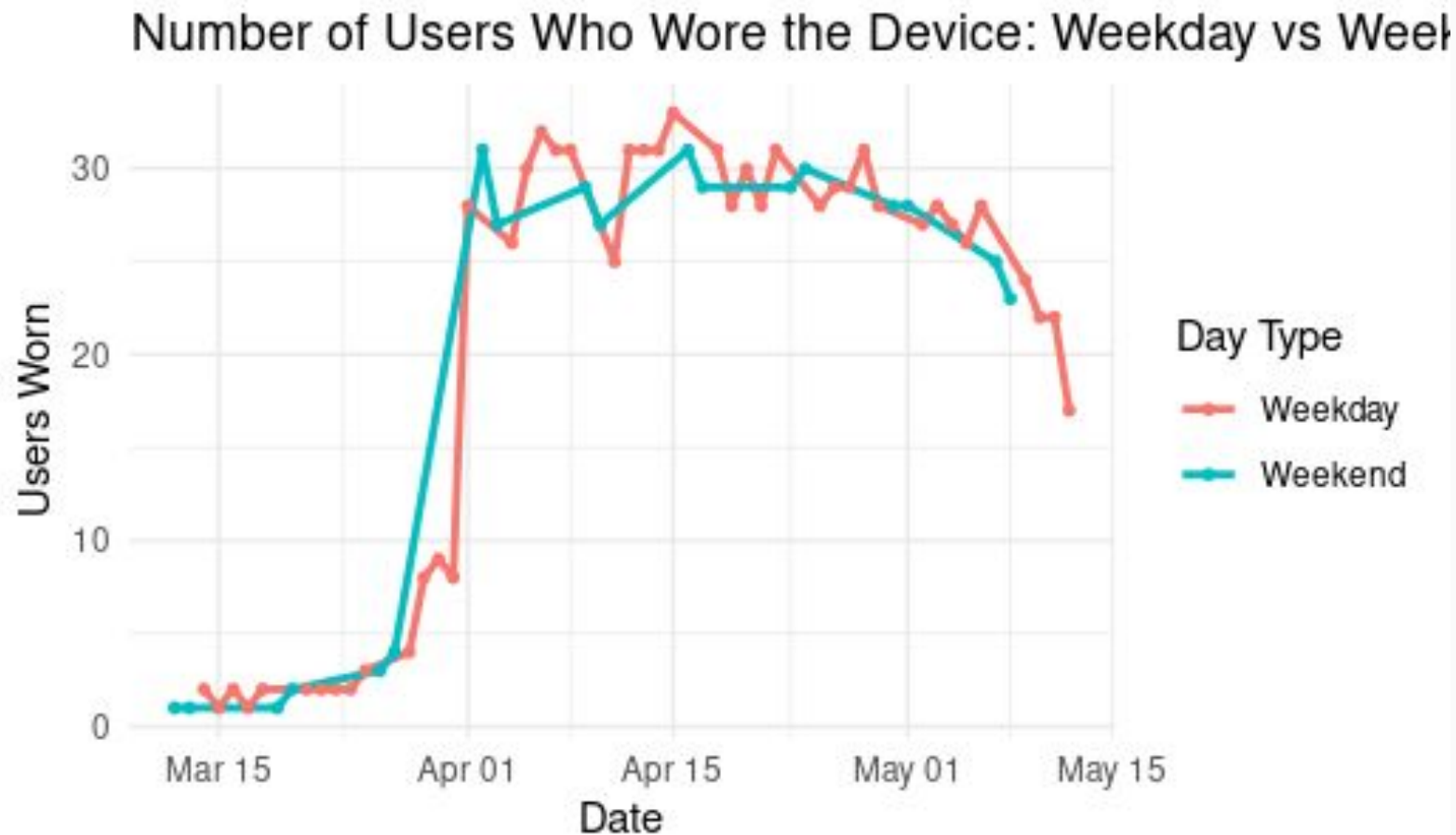
Did consumer activity increase or decrease from April to May?



Do users wear the device consistently across the whole March–May period?



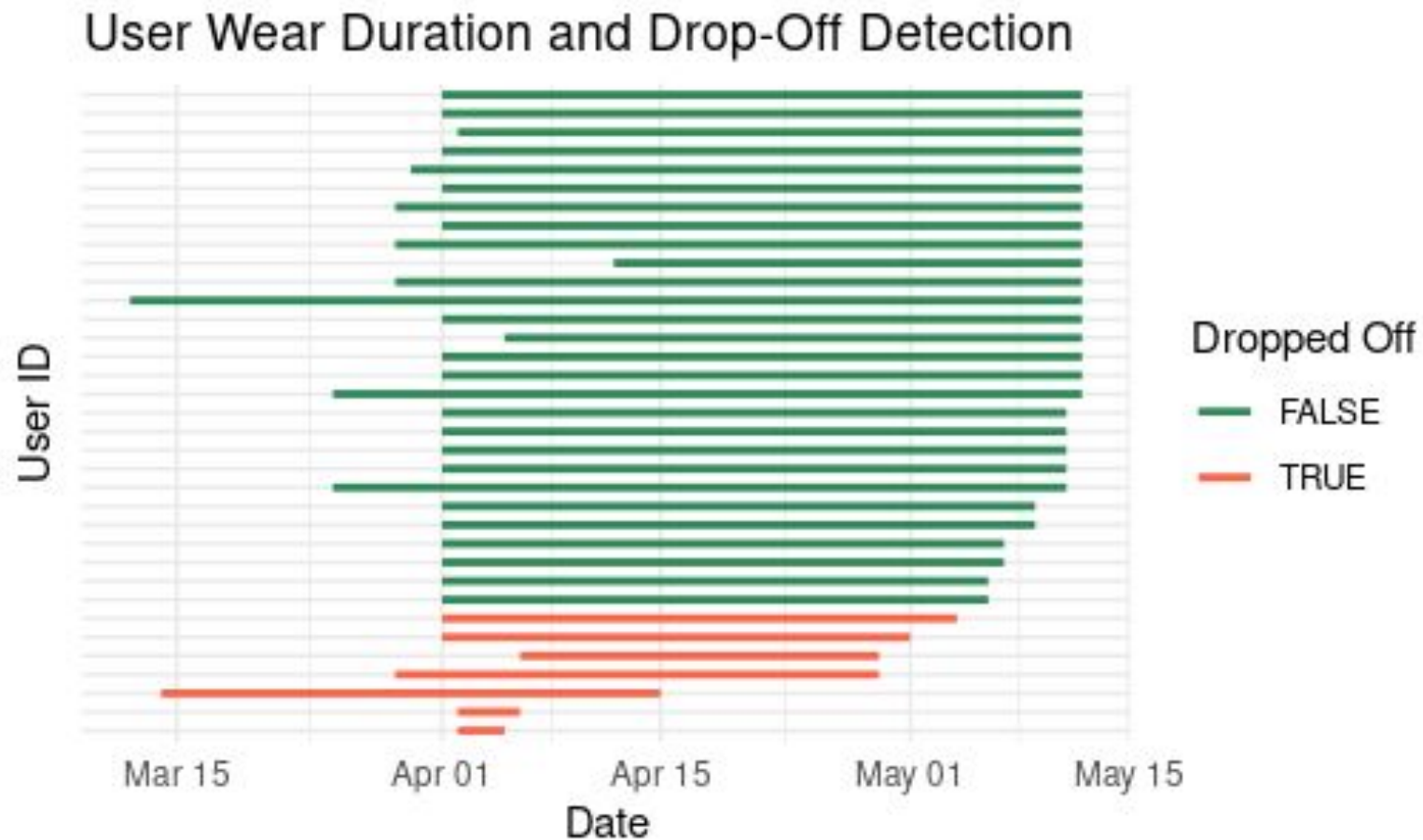
Number of Users Who Wore the Device: Weekday vs Weekend



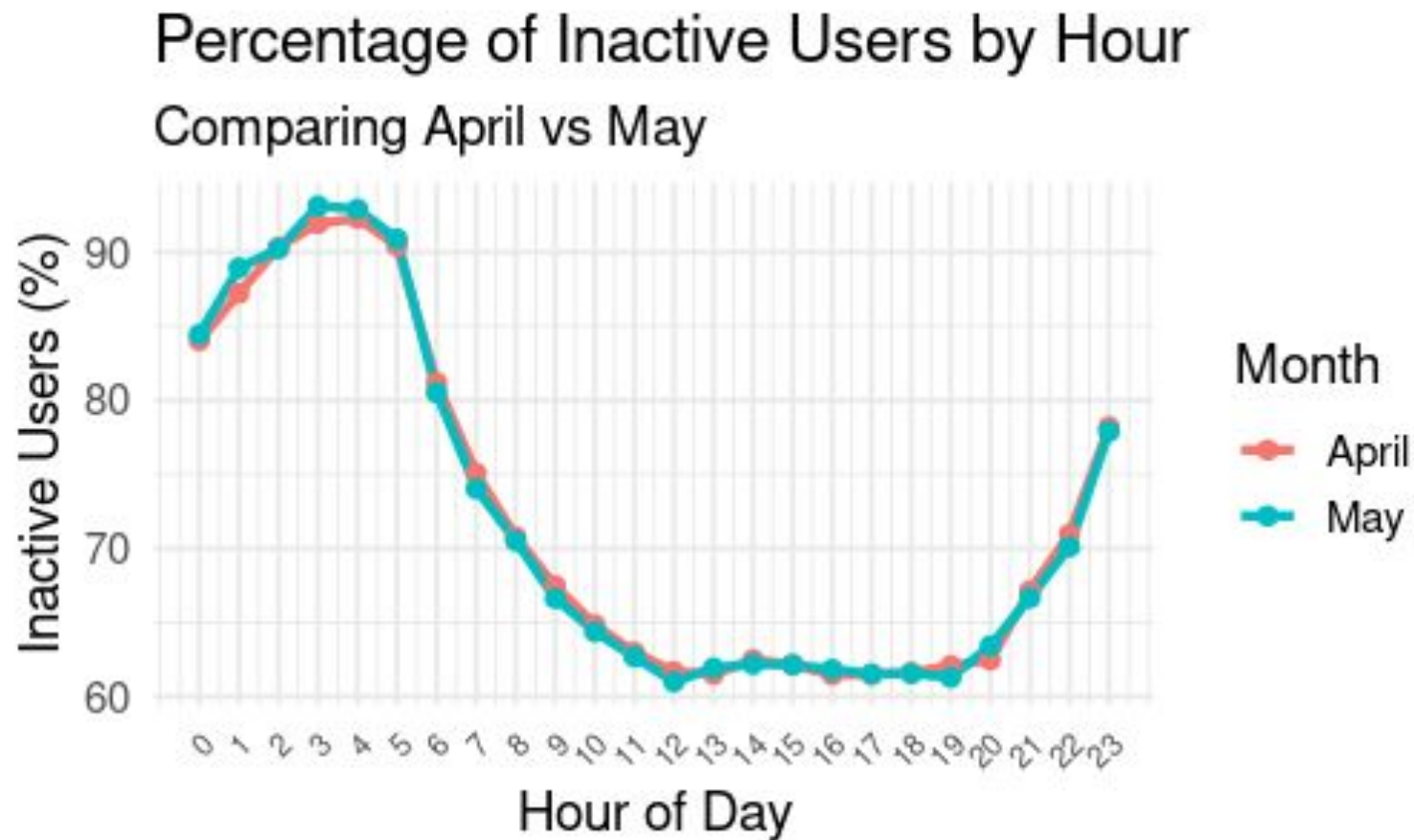
User-level consistency summary



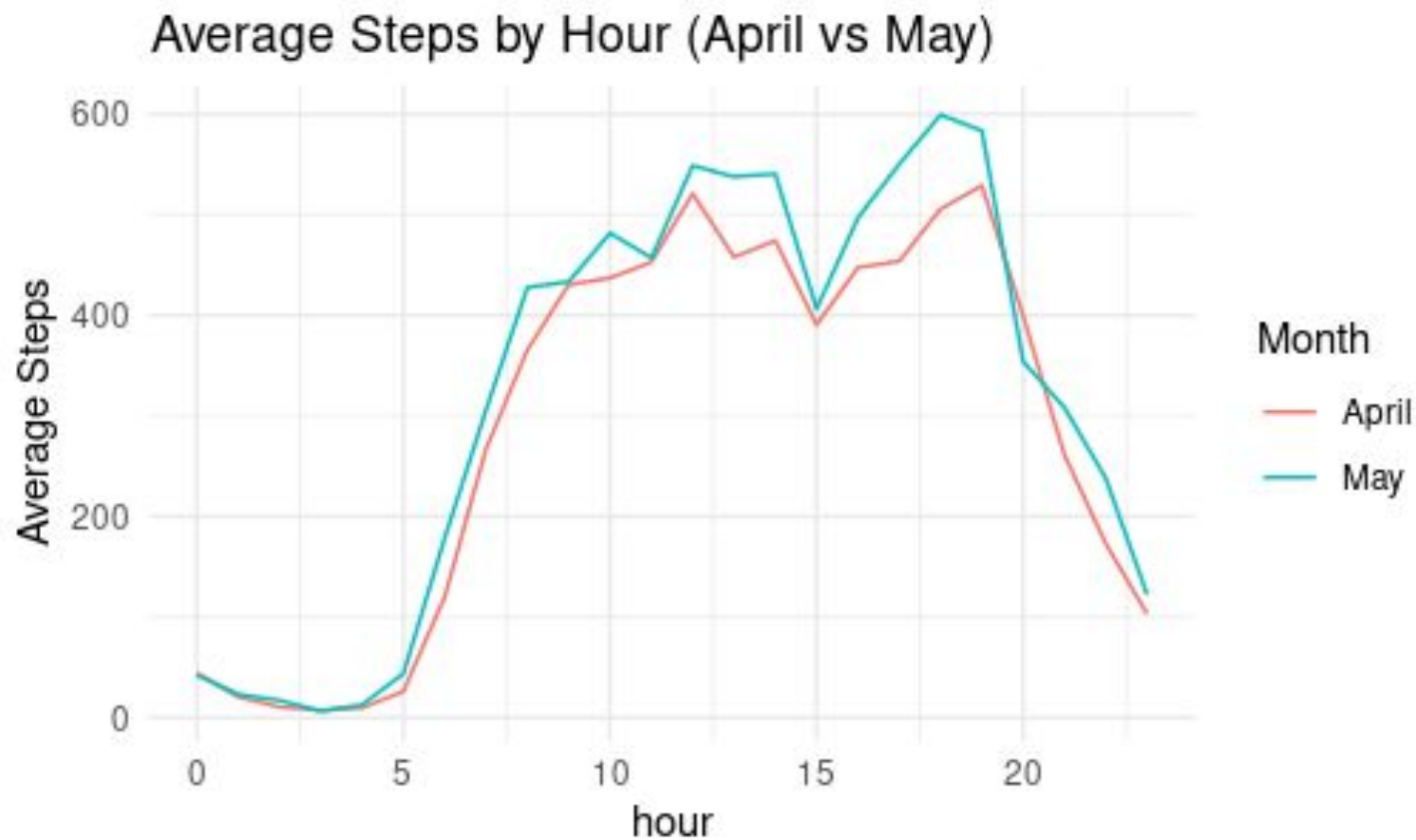
1.4 Drop-off pattern in usage



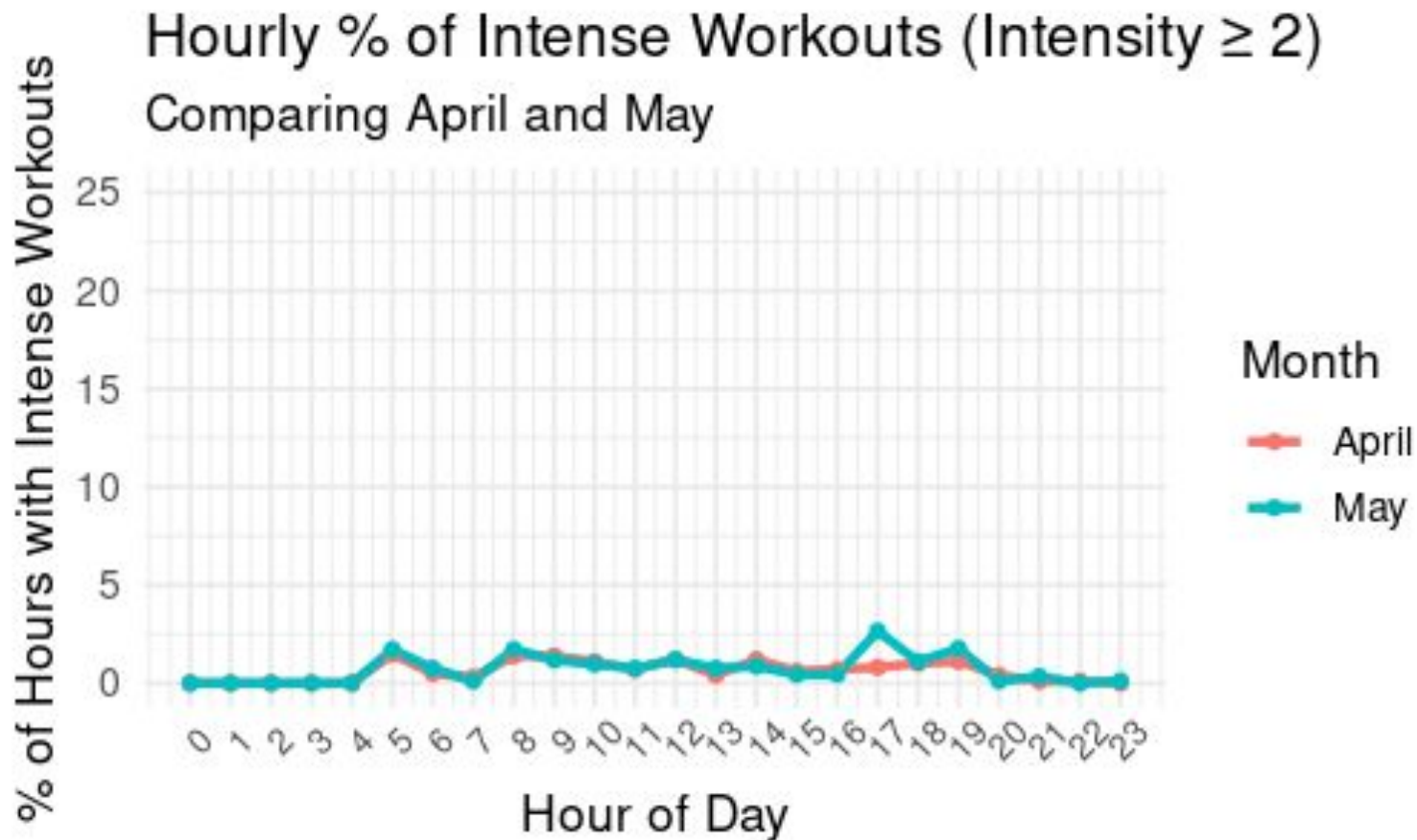
What percentage of users were inactive during each hour (per month)?



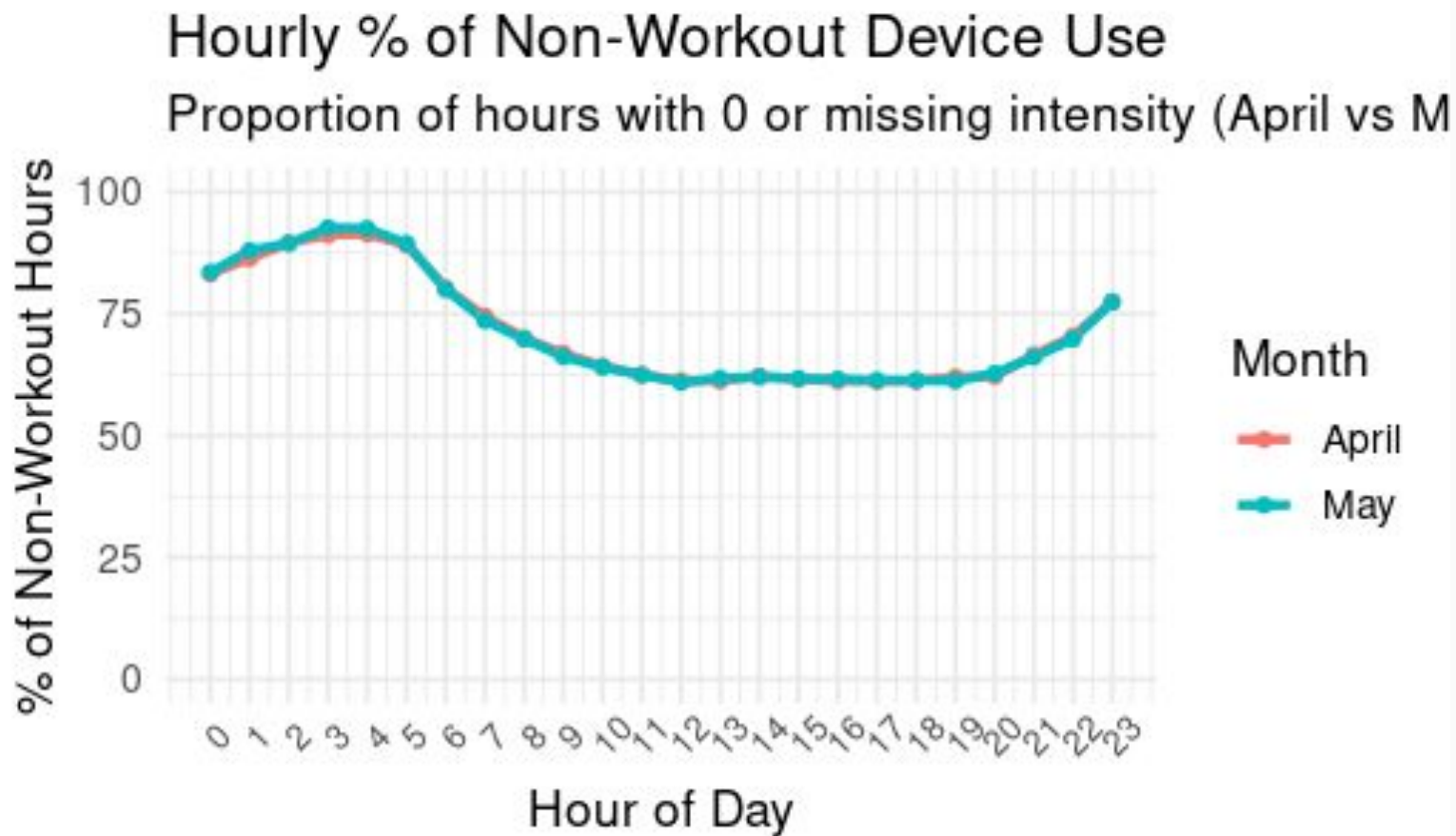
How does step activity differ by hour across months?



Do women work out using the device?

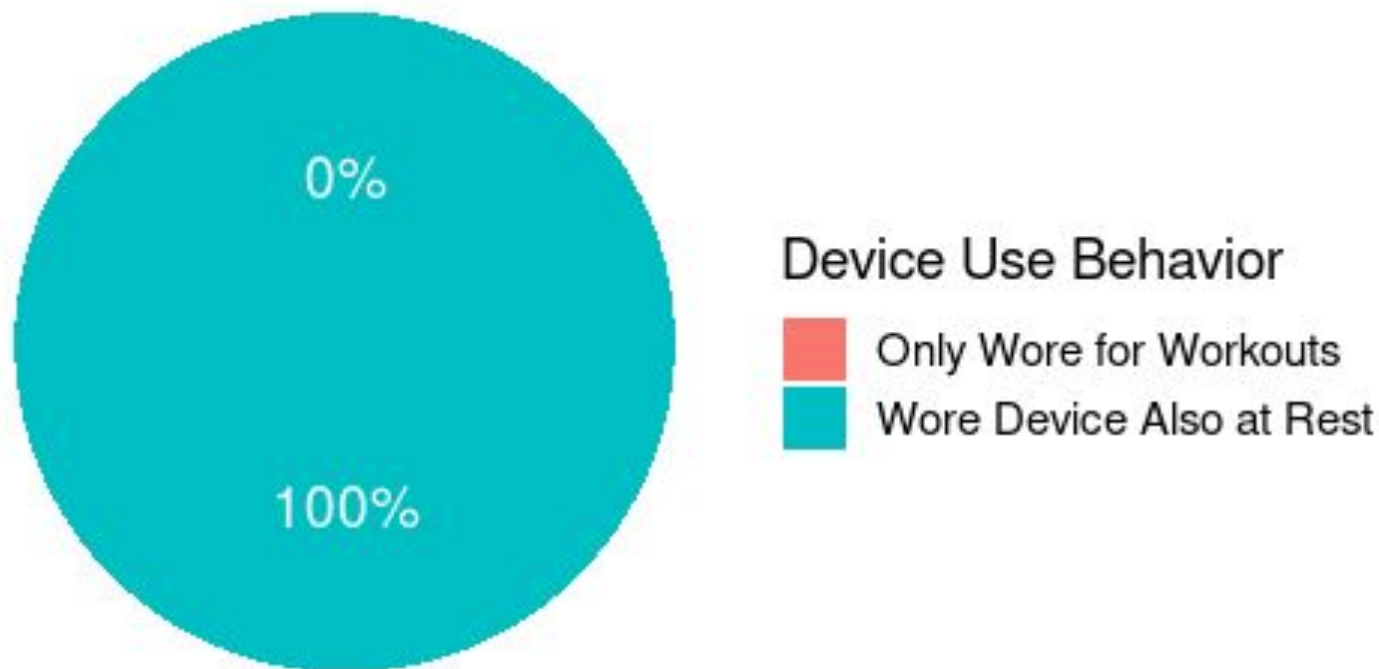


Do women work out using the device?



Do the users who work out ever wear the device outside workout times?

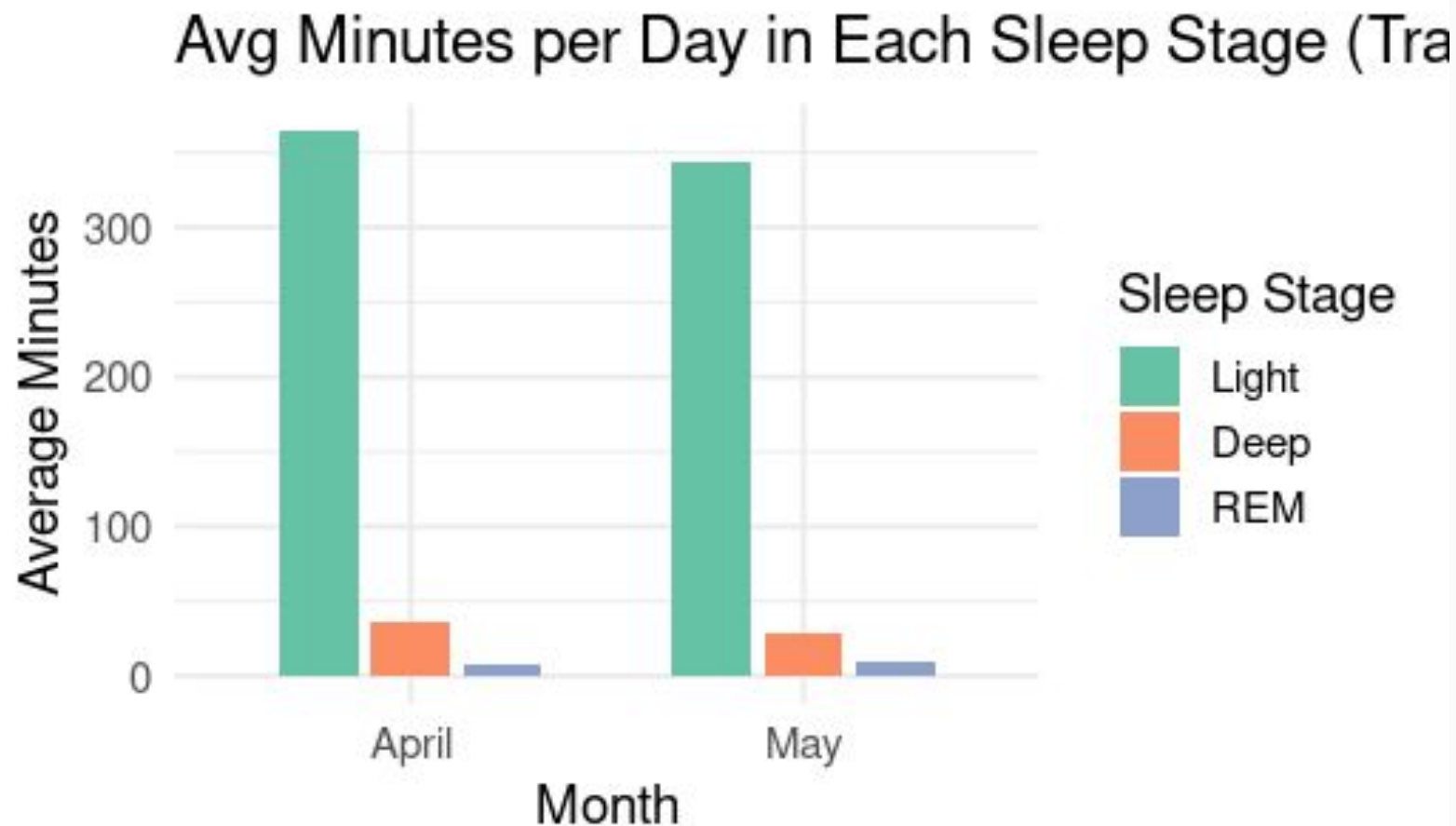
Among Women Who Work Out, Do They Also Wea



Do women wear the device during sleep?



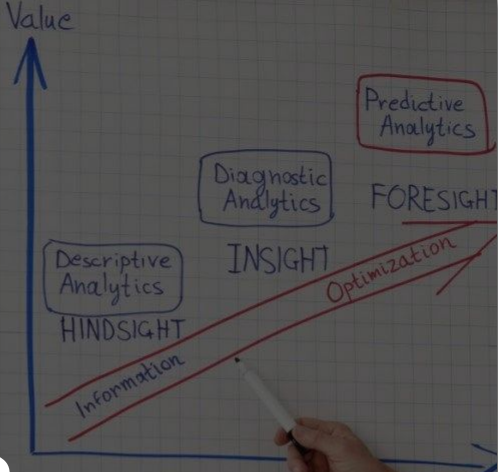
Do women monitor sleep routines?



Recommendations



Insights



- Most users wore the device regularly
- Slight drop-off near the end of the study
- Over 60% used it consistently week to week
- Peak activity: 8 AM – 8 PM
- Used for both workouts and everyday tracking
- Step counts changed from April to May (some up, some down)
- No big difference between weekdays and weekends
- Sleep tracking was inconsistent (many skipped it)
- Those who wore it overnight captured meaningful sleep data

Recommendations

- Kickstart with onboarding to boost early adoption
- Use reminders & streak rewards to reduce drop-off
- Promote benefits of regular, all-day use
- Send prompts during peak hours (8 AM – 8 PM)

- Encourage overnight wear with comfort tips
- Re-engage users with step drop-offs (April–May)
- Keep weekday/weekend messaging consistent, with light weekend boosts
- Personalize by behavior type: consistent, drop-offs, and sleep skippers

Appendix 1: Data Tables

Question	Table(s) Used
Do women wear the device on a daily basis?	daily_activity_merged
And consistently over time?	daily_activity_merged
Is there a drop-off pattern in usage?	daily_activity_merged
Do women track movement throughout the day?	hourly_steps_merged, hourly_intensities_merged
Do women wear the device during sleep? How often?	sleepday_merged
Do women monitor sleep routines?	minute_sleep_merged
Do usage patterns change on weekends?	daily_activity_merged
Do women work out using the device? Consistently over time?	hourly_intensities_merged
Do women wear the device only or mostly during workouts?	hourly_intensities_merged

Appendix 2: Tools used

Primary Tool: R

- Used R for all stages of the project:
 - Data preparation – importing, formatting
 - Data processing – merging, filtering, parsing, handling missing values
 - Exploratory analysis – identifying usage patterns
 - Visualization – creating charts for steps, sleep, and activity trends



Key packages:

tidyverse, janitor, lubridate, ggplot2, dplyr, readr

Detailed steps and scripts are available in the full report